Lecture 2: Areal Data and Autocorrelation

SISMID 2026

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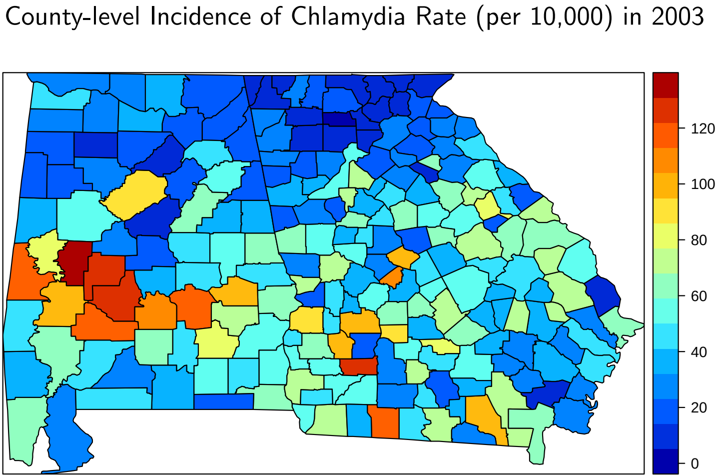
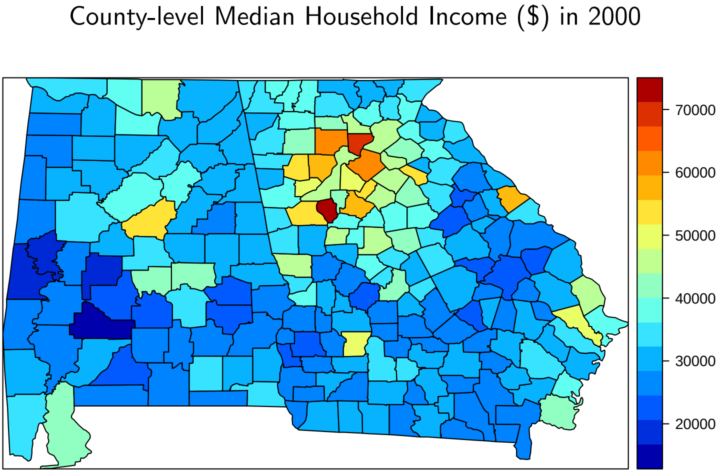
Motivating Data Example



Motivating Data Example



Spatial Analysis Questions

Areal data are typically represented as contiguous polygons with irregular or regular shapes (e.g grid).

Question 1: Is there spatial dependence? Do counties with high median income/chlamydia incidence tend to be closer to each other?

Question 2: What are some covariates that explain the observed spatial pattern? The covariates of interest often exhibit spatial pattern themselves.

Question 3: Can we identify areas that have values much higher or lower than expected? Perhaps these areas require further investigation or intervention.

Question 4: How do we smooth across polygons (leveraging spatial correlation) to account for random noise associated with each observation?

Proximity/Adjacency Matrix **W**

The spatial information of areal data cannot be represented by a single location. We need to describe how each observation is spatially connected to all other areal units.

Given observations $Y_1, \ldots, Y_n$, let **W** denote an $n \times n$ matrix, where element W_{ij} measures the spatial proximity between Y_i and Y_j.

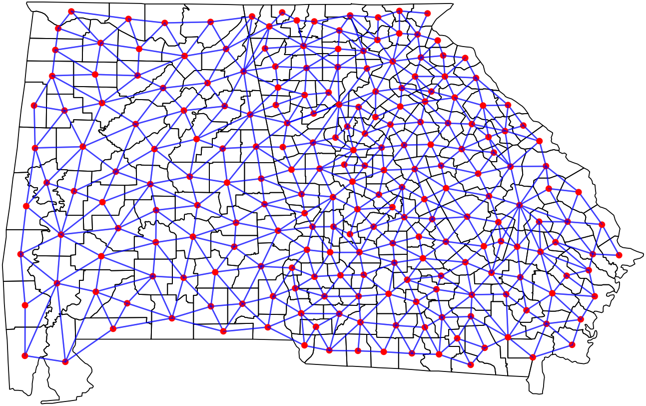
Some common choices include:

- $W_{ij} = 1$ if Y_i and Y_j share a border and $W_{ij} = 0$ otherwise.
- $W_{ij} = d_{ij}$ where d_{ij} is the distance between the centroids of area i and area j.
- $W_{ij} = 1$ if d_{ij} is below a threshold and $W_{ij} = 0$ otherwise.

Note that the above three choices give us a symmetric **W**. This not is a requirement. One example of an asymmetric **W** is when $W_{ij} = 1$ if Y_j is one of the K closest areas to Y_i.

Also, by definition, we have $diag(\mathbf{W}) = \mathbf{0}$.

Counties Sharing a Border

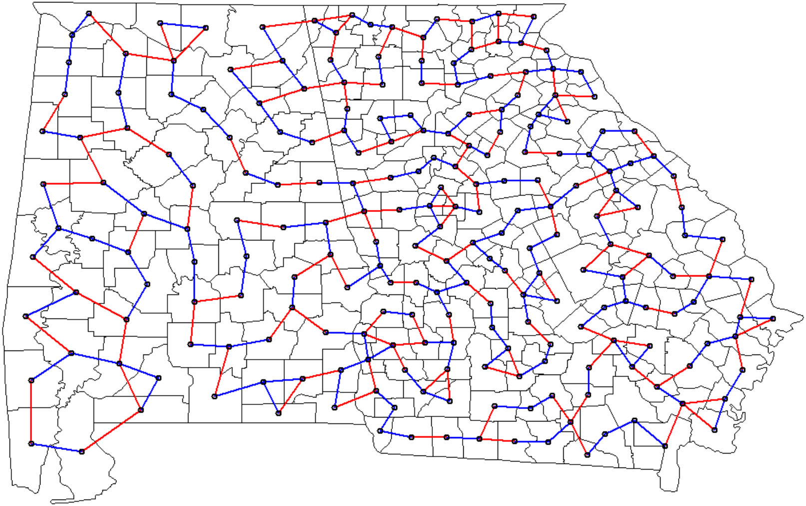


Number of regions = 226
Number of links = 1,258
Average number of links = 5.6

First- or Second-closest County

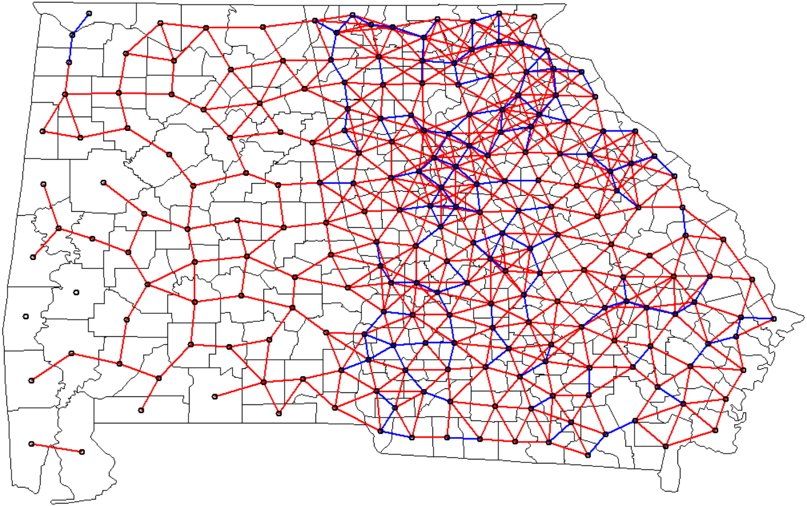
Blue = first closest county

Red = second closest county



Distance Buffer

Blue = centroids within 25km Red = centroids within 40km



Measure of Spatial Dependence

Spatial dependence implies that each Y_i is correlated with its neighbors. Consider the following regression model

$$Y_i = \rho \times \text{Average of neighbors of } Y_i + \epsilon_i.$$

A positive/negative ρ value indicates positive/negative spatial dependence.

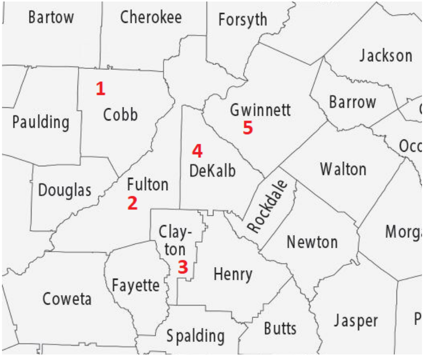
We can specify this regression using adjacency matrix $\mathbf{W}$. Define $\widetilde{\mathbf{W}}$ as a row-standardized (sum to 1) proximity matrix

$$\widetilde{W}_{ij} = W_{ij} / \sum_{i=1}^{n} W_{ij} \ .$$

$$\mathbf{Y} = \rho \widetilde{\mathbf{W}} \mathbf{Y} + \boldsymbol{\epsilon}.$$

- Row-standardization accounts for the different number of neighbors for each polygon.
- $\widetilde{\mathbf{W}}\mathbf{Y}$ is a.k.a. *spatially-lagged* covariate.

An Example: 5-county Atlanta



$$W = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

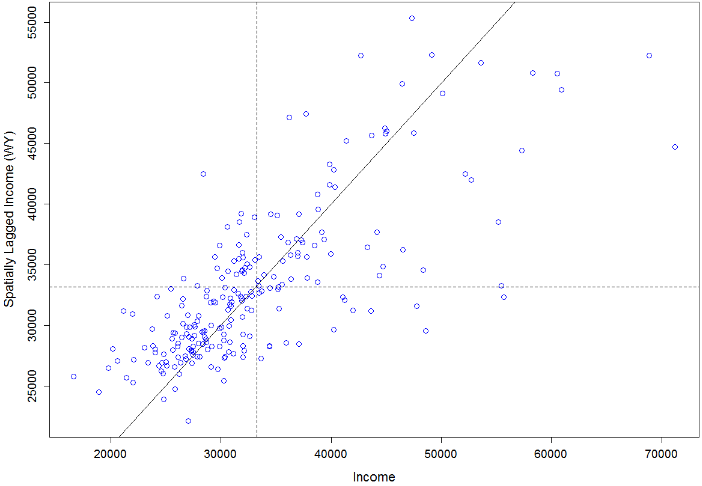
An Example: 5-county Atlanta

$$W = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix} \qquad \widetilde{W} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1/4 & 0 & 1/4 & 1/4 & 1/4 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 1/3 & 1/3 & 0 & 1/3 \\ 0 & 1/2 & 0 & 1/2 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \rho\widetilde{W} = \rho \begin{bmatrix} y_2 \\ (y_1 + y_3 + y_4 + y_5)/4 \\ (y_2 + y_4)/2 \\ (y_2 + y_3 + y_5)/3 \\ (y_2 + y_4)/2 \end{bmatrix}$$

Measure of Spatial Dependence (Moran's Plot)

Spatially-weighted Income ($\tilde{\mathbf{W}}\mathbf{Y}$) versus Observed Income ($\mathbf{Y}$)



Moran's I

Given any proximity matrix **W**, the Moran's I statistics is

$$I = \frac{n}{\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij}} \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij}(Y_i - \bar{Y})(Y_j - \bar{Y})}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}$$

where $\bar{Y}$ is the average of all Y_i's. The function of W_{ij} is to select neighbors for unit i.

Note that if we use the row-standardized proximity matrix $\tilde{\mathbf{W}}$, then

$$I = (\mathbf{Y}'\tilde{\mathbf{W}}\mathbf{Y})/(\mathbf{Y}'\mathbf{Y})$$

which is identical to the slope coefficient from a linear regression of $\tilde{\mathbf{W}}\mathbf{Y}$ on $\mathbf{Y}$.

- $I > 0$ indicates positive spatial dependence.
- $I < 0$ indicates negative spatial dependence.

Moran's I

- Moran's I is unit-less.
- Moran's I depends on the proximity/weight matrix **W**.
- Moran's I has a range given by the minimum and maximum eigenvalue of $(\mathbf{W} + \mathbf{W}')/2$

For the income data, we have a Moran's I of 0.541.

The range of possible Moran's I values using a **W** defined as the all neighbours sharing a common boundary is -0.566 to 1.016.

Moran's I: Hypothesis Test

In exploratory analysis, we are interested in testing whether I is meaningfully larger or smaller than 0.

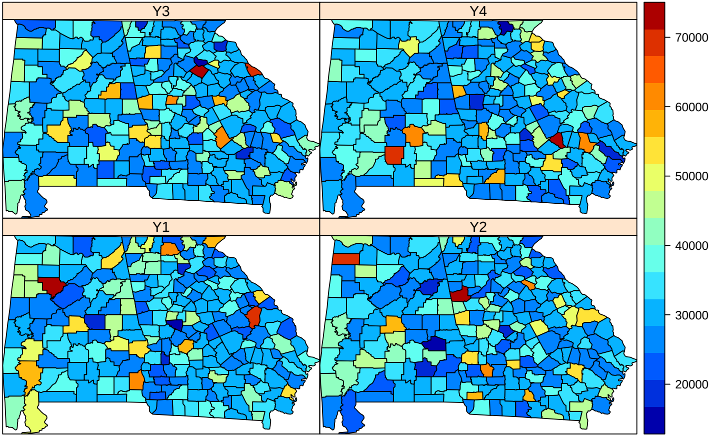
Approach 1: Asymptotic Normality:

- Under the null hypothesis $I = 0$, the expected value of I has mean $-1/(n-1)$ and is asymptotically normal. The standard error is very complicated and depends on whether we assume the outcome Y is Gaussian.
- For our previous income analysis, the Z-value is about 13, indicating strong evidence for the presence of spatial autocorrelation.

Approach 2: Permutation Test:

- The null hypothesis implies that the observed Y_i's is independent of their locations.
- The sampling variability in I under the null hypothesis can be approximated by randomly permuting Y_i across the areal units.

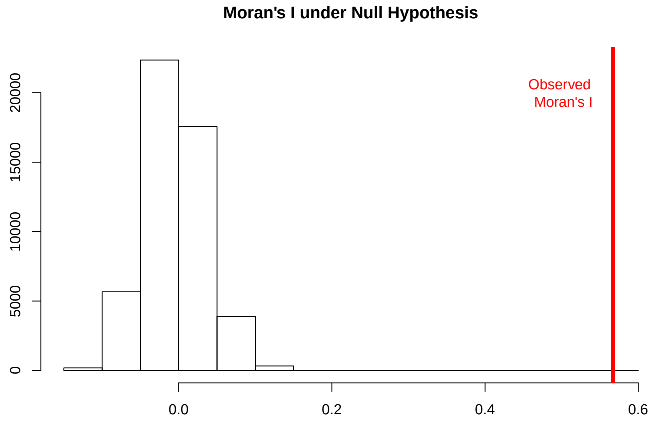
Moran's I: Permutation



The corresponding Moran's I are: -0.02, -0.05, -0.03, 0.02.

Moran's I: Permutation

Permutation p-value $< 1 \times 10^{-4}$ based on 50,000 permutations.



Global versus Local Dependence

The Moran's I statistic measures global spatial dependence because it uses one number of summarize the entire spatial region.

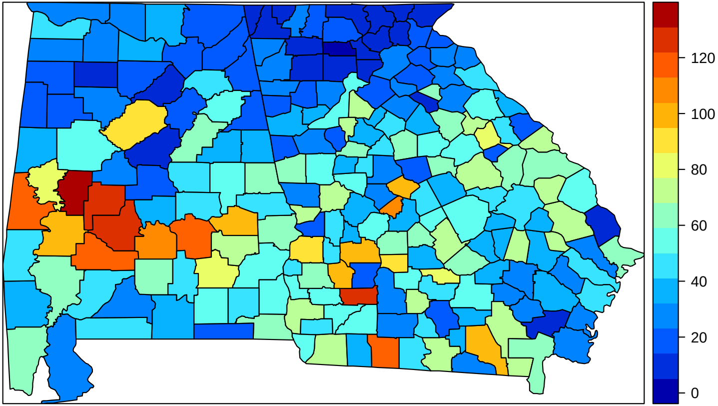
Sometimes *clustering* can occur at spatial sub-regions. A local measure that varies spatially can be useful.

Local Indicator of Spatial Association (LISA)

- The LISA for each observation gives an indication of the extent of significant spatial clustering of similar values around that observation.
- The sum of LISA's for all observation is proportional to a global indicator of spatial association.

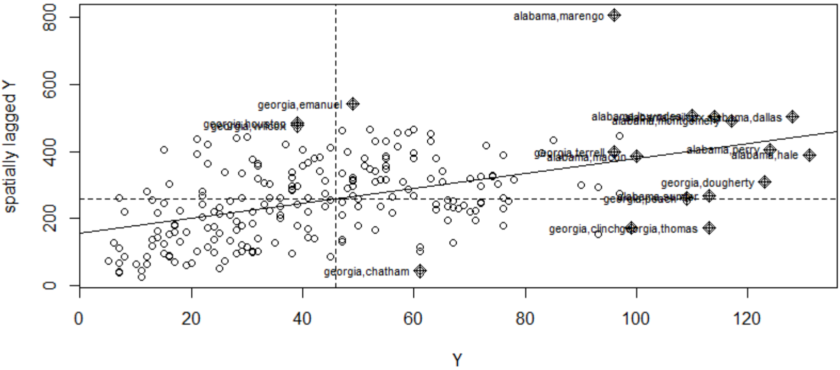
Data Example

County-level Incidence of Chlamydia per 10,000 in 2003



Moran's I Plot for Chlamydia Incidence

Global Moran's I = 0.357
The labeled points have high influence.



Local Moran's I (Anselin 1995)

The local Moran's I statistic, at location i, is defined as

$$I_i = \frac{Y_i - \bar{Y}}{\sum_{k=1}^{n}(Y_k - \bar{Y})^2/(n-1)} \sum_{j=1}^{n} W_{ij}(Y_j - \bar{Y}) \quad .$$

- I_i measures how different Y_i is from its spatial weighted average.
- I_i is scaled by the total variability $s^2 = \sum_k (Y_k - \bar{Y})^2/(n-1)$.

I_i is large when

- $Y_i - \bar{Y}$ is large $\rightarrow$ Y_i is different from the average value.
- $\sum_j W_{ij}(Y_j - \bar{Y})$ is large $\rightarrow$ Y_i's neighbor average is different from the average.

I_i is positive = clustering; I_i is negative = repulsion.

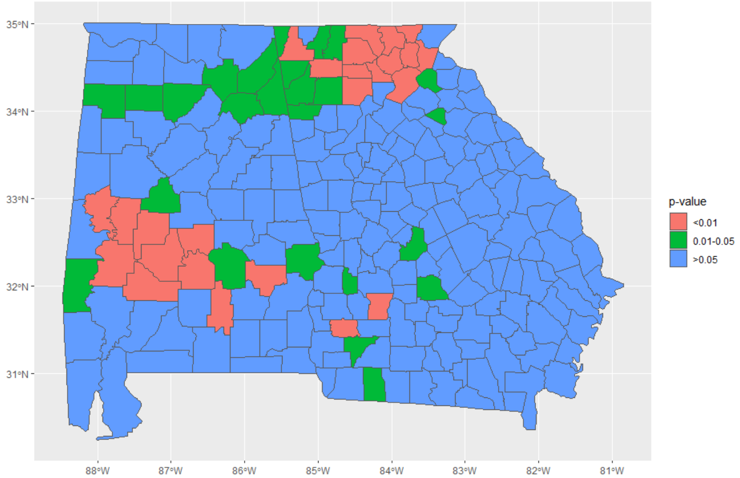
Local Moran's I

Note how the Global Moran's I can be decomposed.

$$
\begin{aligned}
I &= \frac{n}{\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij}} \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij}(Y_i - \bar{Y})(Y_j - \bar{Y})}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2} \\
&= \left(\frac{n}{\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij}} \right) \sum_{i=1}^{n} \left[\frac{(Y_i - \bar{Y})}{\sum_{k=1}^{n}(Y_k - \bar{Y})^2} \sum_{j=1}^{n} W_{ij}(Y_j - \bar{Y}) \right] \\
&= \text{Constant} \times \sum_{i=1}^{n} I_i \ .
\end{aligned}
$$

- Sum of I_i is proportional to the global Moran's I.
- For each I_i, we can perform hypothesis test using its asymptotic variance.

Cluster Detecting with Local Moran's I



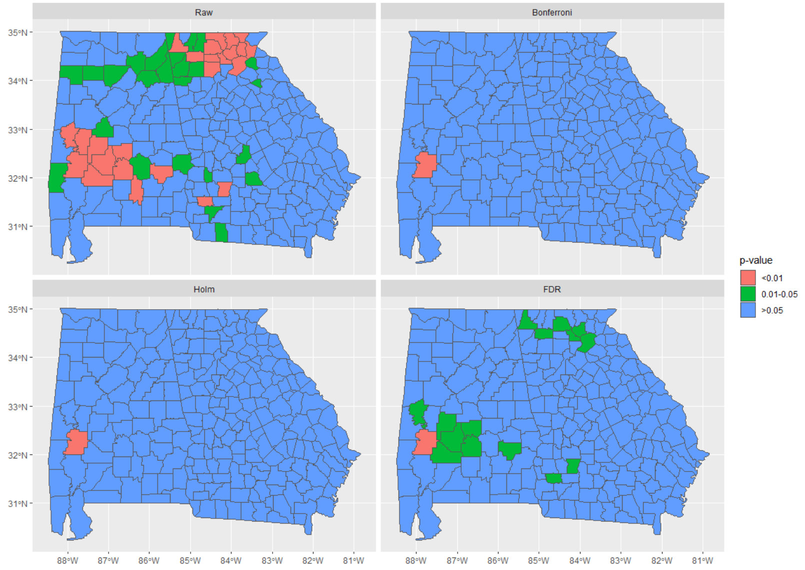
Control for Multiple Testing

For cluster detection, a hypothesis test is performed at each spatial unit. To protect against inflated type I error rate, a multiple testing correction is often implemented.

Let p_i and $\tilde{p}_i$ be the original and the corrected p-value, respectively.

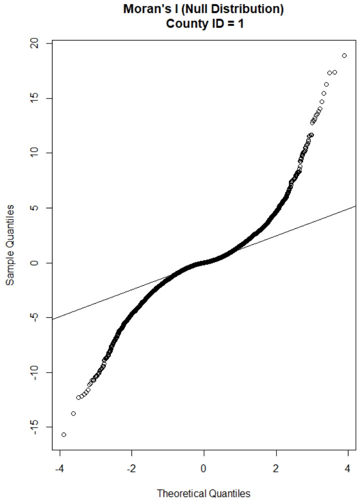
1. Bonferroni
   - $\tilde{p}_i = \min\{1, p_i \times n\}$
   - Can be overly conservative (i.e. true family-wise type I error $<< \alpha$).
2. Holm
   - $\tilde{p}_{[i]} = \min\{1, p_{[i]} * \times(n - i + 1)\}$, where $p_{[i]}$ is the i^{th} smallest p-value.
   - Controls for family-wise type I error rate α but less conservative than Bonferroni (i.e. more power).
3. False-discovery rate (Benjamin-Hochberg)
   - Controls for the expected proportion of falsely rejected H_0 out of total rejections.
   - Less stringent than family-wise type I error (mainly for screening).

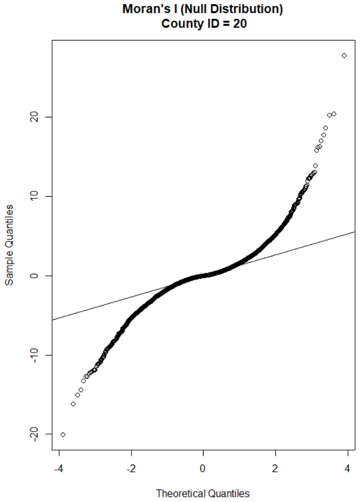
Local Dependence with Local Moran's I



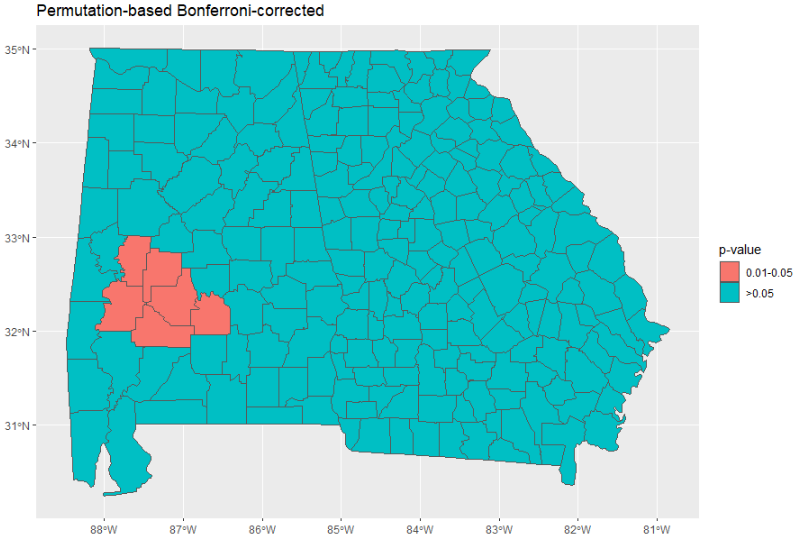
Monte Carlo-Based Inference

Local inference is more prone to normality violation because the neighbors around each spatial unit is often very small.

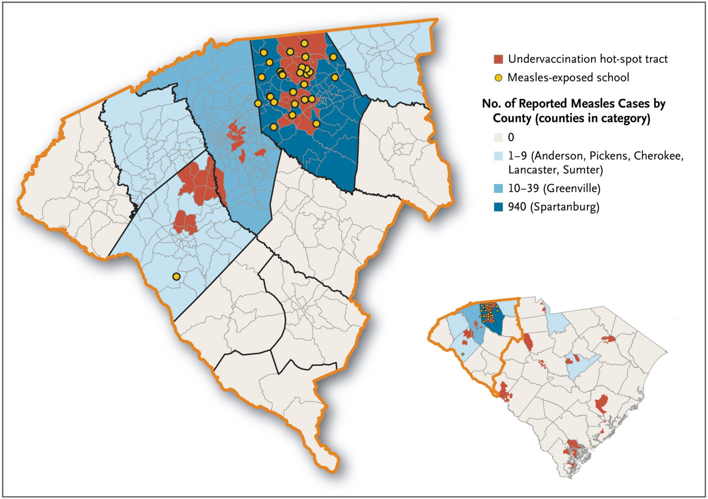




Permutation Test for Local Moran's I



Undervaccinated Hot Spots and Measles Cases



Hot spots identified by Getis-Ord statistics: $G_i = \dfrac{\sum_j W_{ij} Y_j}{\sum_j Y_j}$

Serman EA et al. *NEJM*. 2026 May 20.